# ESE 2150 Lab 4

Op-Amp Comparators and an Intensity Display System

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## A. Op-Amp Power Up and a Simple Comparator

### 1. Measurement Procedure and Observations

First, our team assembled the circuit. We used an ohmmeter to check that the 4.7 kΩ resistors had the correct resistance. We also used a voltmeter, connecting its negative terminal to ground and probing each node with the positive terminal, to check that the circuit was wired correctly. The LM324AN was powered using $V_{CC+} = 5\,\text{V}$ and $V_{CC-} = -5\,\text{V}$.

For our measurements, we connected the voltmeter's negative terminal to the circuit ground and placed its positive terminal at $V_p$, the non-inverting input of the op-amp. We adjusted the trimmer until the voltmeter reading matched the desired $V_p$. Leaving the trimmer untouched, we moved the positive terminal to $V_n$ and $V_o$ and measured their respective voltages. We repeated each measurement three times to minimize error and recorded the averages. We then adjusted the trimmer to the next desired voltage and repeated the procedure.

We noticed that $V_n$ stayed close to 2.497 V, while $V_o$ switched from approximately $-4.997\,\text{V}$ to $3.758\,\text{V}$ between the sampled inputs of $V_p = 2.5\,\text{V}$ and $V_p = 3\,\text{V}$. This is consistent with the circuit acting as a comparator. When the non-inverting input is below the inverting input, the output saturates low. When the non-inverting input is above the inverting input, the output saturates high.

Setup photograph

Upload `setup_a.jpg` to insert the image.

Figure 1: Comparator circuit setup for Part A.

## 2. Measurement Results

| Adjusted $V_p$ (V) | Measured $V_n$ (V) | Measured $V_o$ (V) |
|---:|---:|---:|
| 0.0 | 2.497 | -4.997 |
| 0.5 | 2.489 | -4.997 |
| 1.0 | 2.497 | -4.997 |
| 1.5 | 2.497 | -4.997 |
| 2.0 | 2.497 | -4.997 |
| 2.5 | 2.497 | -4.997 |
| 3.0 | 2.497 | 3.758 |
| 3.5 | 2.497 | 3.758 |
| 4.0 | 2.497 | 3.758 |
| 4.5 | 2.497 | 3.758 |
| 4.8 | 2.497 | 3.758 |

Table 1: Measured comparator input and output voltages.

## 3. Plots

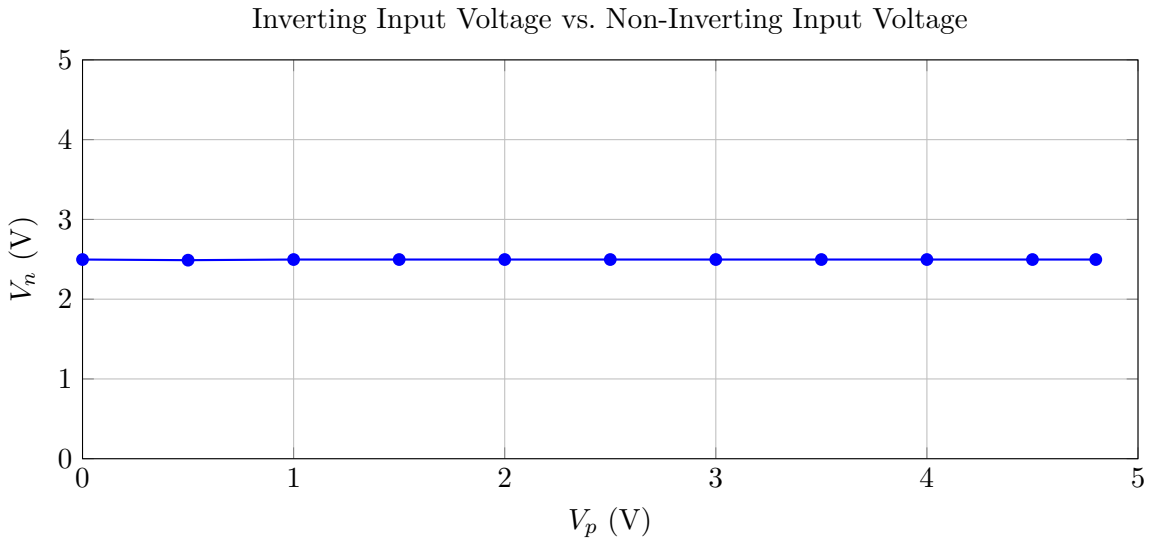



Figure 2: $V_n$ stays approximately constant because its voltage divider depends on the positive supply rather than on $V_p$.

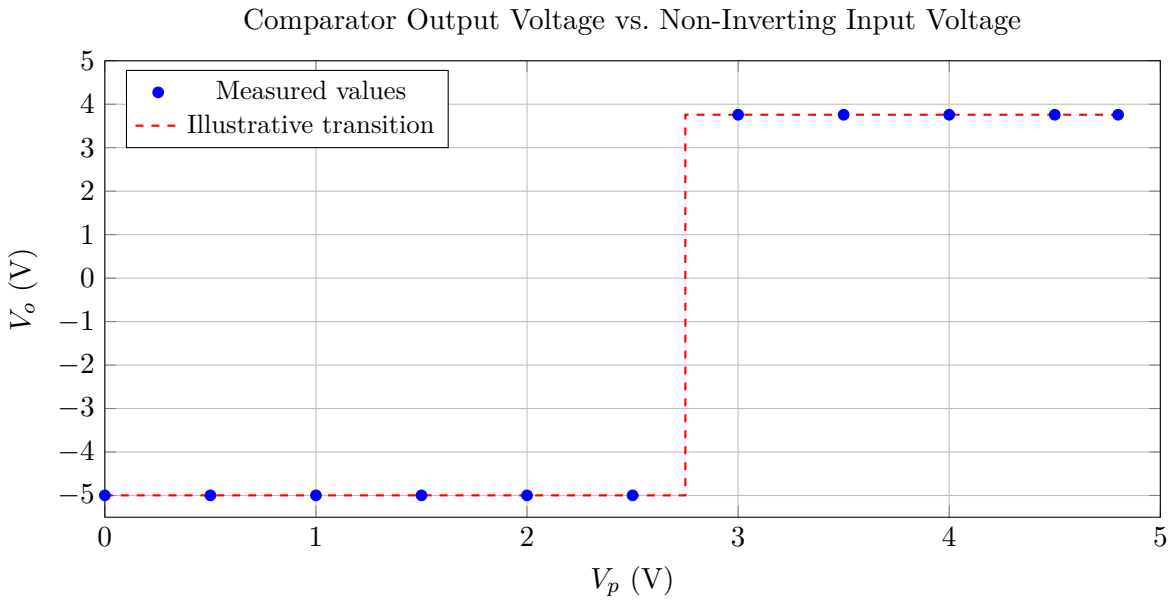



Figure 3: The output switches between low and high saturation. The dashed transition is illustrative; the samples do not establish the exact switching voltage.

### 4. Is the Comparator Response Symmetric?

The comparator response is not symmetric about the horizontal axis. With power supplies of $+5\,\mathrm{V}$ and $-5\,\mathrm{V}$, the measured low output was $-4.997\,\mathrm{V}$, while the high output was $3.758\,\mathrm{V}$.

The low output was only

$$|-4.997 - (-5)| = 0.003\,\mathrm{V}$$

from the negative supply rail. The high output was

$$5 - 3.758 = 1.242\,\mathrm{V}$$

below the positive supply rail.

The LM324AN cannot drive its output all the way to the positive supply under these conditions. Its output stage requires voltage headroom, so the positive and negative saturation levels have different magnitudes even though the supply rails are symmetric.

## B. Comparator in Action: Intensity Display System

### 1. Measurement Procedure and Observations

Our group assembled the intensity display circuit, making sure all components were placed correctly. We used an ohmmeter to check the resistors and a voltmeter to verify the node voltages. We connected the voltmeter's negative terminal to ground and probed each node with the positive terminal.

After checking the circuit, we measured $V_1$, $V_2$, $V_3$, and $V_4$, the inverting inputs of the four comparators. We recorded $0.802\,\mathrm{V}$, $1.604\,\mathrm{V}$, $2.404\,\mathrm{V}$, and $3.207\,\mathrm{V}$, respectively. All four measurements were within approximately 2% of their calculated values.

The reference voltages increased from $V_1$ to $V_4$ because each successive node was farther above ground along the resistor ladder. We then applied a sinusoidal input at 200 mHz and varied its amplitude. Increasing the amplitude caused more LEDs to turn on. At an amplitude of 1 V, only LED 1 turned on; at 4 V and 5 V, all four LEDs turned on during part of the cycle.

Finally, we set the amplitude to 5 V and increased the frequency. The LEDs blinked faster as frequency increased. At 10 Hz, they appeared to flicker together, and at 50 Hz, the flickering became difficult to distinguish visually.

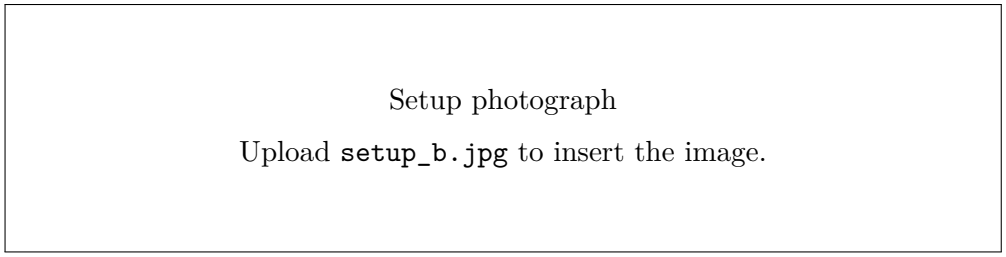



Figure 4: Intensity display circuit and measurement equipment.

## 2. Reference Voltage Calculations

The resistor ladder consists of one 10 kΩ resistor and four 4.7 kΩ resistors in series. Its total resistance is

$$R_{\text{total}} = 10\,\text{k}\Omega + 4(4.7\,\text{k}\Omega) = 28.8\,\text{k}\Omega.$$

Assuming negligible comparator input currents, the ladder current is

$$I = \frac{5\,\text{V}}{28.8\,\text{k}\Omega} \approx 0.17361\,\text{mA}.$$

The reference voltages relative to ground are

$$\begin{aligned} V_1 &= I(4.7\,\text{k}\Omega) \approx 0.81597\,\text{V}, \\ V_2 &= I(2 \times 4.7\,\text{k}\Omega) \approx 1.63194\,\text{V}, \\ V_3 &= I(3 \times 4.7\,\text{k}\Omega) \approx 2.44792\,\text{V}, \\ V_4 &= I(4 \times 4.7\,\text{k}\Omega) \approx 3.26389\,\text{V}. \end{aligned}$$

| Quantity | $V_1$ (V) | $V_2$ (V) | $V_3$ (V) | $V_4$ (V) |
|---|---|---|---|---|
| Measurement | 0.802 | 1.604 | 2.404 | 3.207 |
| Calculation | 0.816 | 1.632 | 2.448 | 3.264 |

Table 2: Measured and calculated reference voltages.

Our measured node voltages were approximately 1.7% to 1.8% below the calculated values at every node. Because the fractional difference was similar across the ladder, one possible explanation is that the supply voltage was slightly below 5 V. Resistor tolerances and measurement uncertainty could also contribute.

## 3. LED Response Versus Signal Amplitude

| Signal amplitude | 1 V | 2 V | 3 V | 4 V | 5 V |
|---|---|---|---|---|---|
| LED 1 | On | On | On | On | On |
| LED 2 | Off | On | On | On | On |
| LED 3 | Off | Off | On | On | On |
| LED 4 | Off | Off | Off | On | On |

Table 3: LED response at 200 mHz. "On" means that the LED illuminated during some portion of the input cycle, rather than remaining on continuously.

The LEDs turned on sequentially from LED 1 through LED 4 as the input voltage increased. As the input decreased, they turned off in reverse order, from LED 4 through LED 1.

Each comparator has a different positive reference voltage, so its LED turns on only when the input exceeds that threshold. For a sufficiently large sinusoidal input,

$$v_{\text{in}}(t) = A\sin(2\pi f t),$$

the rising waveform crosses $V_1$, $V_2$, $V_3$, and $V_4$ in that order. The falling waveform crosses them in the opposite order.

There was a time gap between sequences because the input spends part of each cycle below the lowest threshold. During the negative half-cycle, none of the positive thresholds are exceeded, so all four LEDs are off.

The time intervals depend on frequency. At 200 mHz, the period is

$$T = \frac{1}{0.2} = 5\,\text{s}.$$

This makes the individual transitions and the gap between sequences much easier to see than at 1 Hz or 10 Hz.

## 4. Frequency Dependence at a 5 V Amplitude

At a 5 V amplitude, the LEDs visibly lit in sequence at 200 mHz, 500 mHz, and 1 Hz. The sequence became faster as the frequency increased.

At approximately 10 Hz, the LEDs appeared to turn on and off together, although rapid flickering was still visible. At 50 Hz, they appeared nearly continuously lit because the cycle repeated too quickly for us to clearly distinguish the individual transitions.

For example, at 10 Hz, the period is

$$T = \frac{1}{10} = 0.1\,\text{s} = 100\,\text{ms}.$$

Using the calculated thresholds, the time between LED 1 and LED 4 turning on during a rising positive half-cycle is approximately

$$\Delta t = \frac{\sin^{-1}(V_4/5) - \sin^{-1}(V_1/5)}{2\pi f}.$$

At $f = 10\,\text{Hz}$,

$$\Delta t \approx \frac{\sin^{-1}(3.264/5) - \sin^{-1}(0.816/5)}{2\pi(10)} \approx 8.7\,\text{ms}.$$

The LEDs are still switching sequentially, but the interval is too short for us to easily distinguish by eye. At 50 Hz, the full cycle lasts only

$$T = \frac{1}{50} = 20\,\text{ms},$$

making the separate transitions even harder to resolve. This apparent simultaneous behavior is due to visual perception rather than the LEDs or comparators failing to switch quickly enough.

## 5. Debugging Observation

We had to explicitly connect the op-amp's power supply pins to $V_{CC+}$ and $V_{CC-}$. These supply connections were not shown in the simplified intensity display schematic, but they were necessary for the LM324AN to operate.